Virtual University of Pakistan

Bt301

Introduction to biotechnology

For final term

For final term

- pSC101 contain ------resistance
  Ampicillin
  Tetracycline
  None
- pBR322 is an example of -----constructed cloning vehicle
  In vivo
  In vitro
  Artificially
- pBR322 contains the resistance gene against
  Ampicillin
  Tetracycline
  Both
  None

- in pBR322 replicon element is
  pBR11
  pMB1
  RSf2124
  PstI
- pBR322 have -----target sites for different restriction enzymes
  30
  20
  40
  50
- pBR322 have high copy number and -----weight
  High
  Low
- pBR325 encodes ----resistance in addition to those resistance have psc101 plasmid.
  Ampicillin
  Chloramphenicol
  Tetracycline
- pAT153 another derivative of
  pBR325
  pBR322
  psc101
- plasmids which loss during cell division is called
  High copy number plasmid
  Low copy number plasmid
  Runaway plasmid
  Lambda phage vector
- Wild type is not suitable as a cloning vector because it has ----restriction enzymes sites
  Very low number of
  Too many
  A few
- Charon4A is a replacement vector is this statement
  True
  False
- Charon 60 is a replacement vector is this statement
  True
  False
- Charon 60 has ----- restriction site
  1
  2
  3
  Many
- Charon 4A and 16 are both derivatives of ------
  Bacteriophage
  Lambda

Ecoli

- Charon 60 can be cloned
  2 kbp
  3kbp
  4kbp
  5kbp
- lacZ gene encode for enzyme
  PAL
  4CA
  βGal
  PPR
- first step in the cloning is
  Cutting vector
  Isolation of DNA
  Insertion of vector
  Infection
- in case phage-λ Packaging yields about ----- of vector DNA
  10^4 plaques/ µg
  10^5 plaques/ µg
  10^6 plaques/ µg
- At each cos site nicks are introduce ----- far apart on opposite strand of DNA
  10bp
  13bp
  12bp
  16bp
- Cos sits are
  Cloning site
  Cooperative site
  Cohesive site
  Capsulation site
- Gene D is include in the --- process
  Capsulation
  Ligation
  Termination
  Packaging
- M13 is a filamentous bacteriophage with-----
  dsDNA
  ssDNA
  dsRNA
  ssRNA
- Which of the following vector replicate without killing its host
  Phage-λ
  M13
  Mp18

Cosmid

- M13 can clone DNA of variable length about
  3kpb
  4kbp
  5kbp
  6kbp
- In x gal containing medium blue colonies represent the cloned DNA colonies, statement is
  True
  False
- Cosmid can clone around ------length of DNA
  45kbp
  37kbp
  53kbp
  200kbp
- Modified cosmid vector produced by ----- in 1981
  Watson and crick
  Ish-Horowicz and Burke
  Edward Southern
  Bates and Swift
- A modified procedure for cosmid is derived by
  Watson and crick
  Ish-Horowicz and Burke
  Edward Southern
  Bates and Swift
- Cosmid c2XB is derived by
  Watson and crick
  Ish-Horowicz and Burke
  Edward Southern
  Bates and Swift
- -----vectors are combination of plasmid and ʎ phage sequences
  Lambda
  Phasmid
  Cosmid
- BACs can clone ----------length of DNA fragment
  5kbp
  150-200kbp
  100-300 kbp
  60kbp
- YACs vectors replicate in ------like normal chromosome
  Bacteria
  Yeast
  Fungi
  Mammalian cell
- YACs can clone ----------length of DNA fragment

  5kbp
  150-200kbp
  100-300 kbp

60kbp

- Vectors that can replicate and are stably maintained in two or more unrelated host organisms are called
  YACs
  Shuttle vectors
  BACs
  Cosmid vector
- Expression vector are used to control
  Infection
  Expression
  Translation
  Insertion
- In Ecoli p1 phage can clone -----length of DNA
  10-15 kb
  2-25 kb
  35-45 kb
  70-100 kb
- The second strategy is to selectively amplify the target sequence directly from the source DNA by using
  Different vectors
  PCR
  Gel electrophoresis
- ---- represent the genomic makeup of an organism
  cDNA library
  RNA sequence
  Genomic library
- Genome size of human is
  $2.8x10^6$
  $3x10^6$
  $1.25x10^5$
- Drosophila melanogaster have size of DNA $1.8x10^5$ this statement is
  True
  False
- Average fragment size can be controlled by----------
  Restriction endonucleases
  Mechanical shearing
  Random cutting
- High molecular weight genomic DNA is digested with-------
  BamH1
  ***Sau*3AI**
  Sal1

PmB1

- For construction of genomic libraries ----vectors are required
  High capacity cloning
  Low capacity cloning
  Moderate capacity cloning
- ------is prepared by reverse-transcribing cellular RNA
  DNA
  cDNA
  Proteins
  Amino acids
- Which lack the non-coding sequence
  DNA
  rRNA
  cDNA
- Introns are rare in bacteria but occur in genes of higher
  Eukaryotes
  Prokaryotes
- cDNA library is representative of the ---population from which it was derived
  DNA
  RNA
  Proteins
- A serious disadvantage of the hairpin method is that cleavage with S1 nuclease results in the loss of sequences at the
  3' end of the clone
  5' end of the clone
- correct orientation of inserted cDNA can be achieved by------
  Hairpin method
  Self-priming method
  Sequencing
  Reverse transcriptase method
- -----followed by the PCR (RTPCR) leads to the amplification of RNA sequences in cDNA form
  Genomic library
  Reverse transcription
  Splicing
  Digestion of DNA
- Major screening strategies involve Genetic methods
  Sequence-dependent screening
  Screening expression libraries
  All of above
- All useful vector molecules carry a selectable ---------or property
  Nutritional marker
  Genetic marker
  Drug resistance
- In phage vectors, ------is itself the selected property

AP^R
TC^R
Plaque formation
CH^R

- Nucleic acid ----------is the most commonly used method of library screening
  Amplification
  Sequencing
  Hybridization
  Screening
- --------- (1975) developed a screening procedure to detect DNA sequences in transformed colonies
  Ish-Horowicz and Burke
  Edward Southern
  Bates and Swift
  Grunstein and Hogness
- The results of the hybridization can be monitored by
  PCR
  Autoradiography
  Gel documentation
  Centrifuge
- Benton and Davis (1977) devised a method called
  Hybridization
  Plaque lift
  Sequencing
  Screening
- In plaque lift techniques there is direct contact between plaque and filter this statement is
  False
  True
- A great advantage of hybridization for library screening is that it is extremely
  Important
  Versatile
  Difficult
  Unique
- A cloned genomic fragment must be found as a ------point for the walk
  Mid
  End
  Starting
  Selectable
- In chromosome walking, human genome starting point may be a ----
  CAPs
  SNPs
  RFLP
  AFLP

- One----- of chromosome walking is the requirement that each DNA segment used is not

repeated elsewhere in the genome
Benefit
Drawback
Characteristics

- In chromosome jumping, the DNA of interest is identified, cut into fragments with restriction enzymes and
  Elongate
  Circularized
  Break down
  Straighten
- Cloned DNAs from the closure sites make up a jumping library is this statement
  False
  True
- The PCR is widely used to isolate specific DNA sequences from-----genomic DNA or cDNA
  Cloned
  Uncloned
  Gel treated
- To isolate specific clone, PCR is carried out with
  Simple primers
  Unusual primers
  Special primers
  Gene-specific primers
- A degenerate primer is a mixture of primers, all of similar sequences but with variations at --- positions
  One
  Two
  One or more
  Several
- If DNA library is established using expression vectors, each individual clone can be expressed to yield a
  Monopeptide
  Polypeptide
  Dipeptide
- Immunochemical screening involves the use of
  Antibodies
  Antigen
  Nucleic acid
  RNA
- The molecular target for recognition is generally an
  Antibody
  Epitope
  Probe
  Primer

- polyvinyl and that IgG antibodies can be readily labelled with
  I^{125}
  I^{108}
  I^{127}

I^{129}

- it is much more convenient to use--- because they have high capacity and efficiency to in vitro packaging
  Bacteriophage simple insertion vector
  Bacteriophage Ecoli 1 insertion vector
  Bacteriophage-ʎ insertion vector
  Bacteriophage-ʎ deletion vectors
- Screening is carried out without using an antibody, by incubating the membranes with radiolabelled ----------
  Ss DNA probe
  ds DNA probe
  Ss RNA probe
- -----is the process by which a particular DNA sequence compensates for a missing function in a mutant cell and thus restores the wild type phenotype
  Functional complementation
  Functional analysis
  Functional genomics
  Non Functional complementation
- Functional complementation is also possible in
  Transgenic animals and plants
  Transgenic animals
  Transgenic plants
  Transgenic bacteria
- Functional complementation used for complementation in -----to isolate the Shaker - 2 gene
  Transgenic cat
  Transgenic mice
  Transgenic bacteria
- -----depends upon transcription of appropriate gene, efficient translation of mRNA and in many cases, posttranslational processing and compartmentalization of nascent polypeptide
  Synthesis of functional DNA
  Synthesis of functional RNA
  Synthesis of functional protein
- Gram-negative bacteria such as E. coli have a
  Complex wall-membrane mitochondria
  Simple wall-membrane
  Complex wall-membrane
  Difficult wall-membrane
- The bacterial inner membrane, periplasmic space and outer membrane all contain proteins found in the cytoplasm
  True

  False
- ABC pathway of protein export and type II secretion pathway are use for
  Chromosome trafficking
  DNA trafficking
  RNA trafficking
  Protein trafficking
- In E.coli foreign protein are

Stable
Instable
Dissolved
Submerged

- In the case of somatostatin, degradation was prevented by producing a ------consisting of somatostatin and β-galactosidase
  Simple protein
  Fused protein
  Complex protein
  Hybrid DNA
- Expression from a strong promoter can represent ----of cloned gene product of toral cell protein
  30-40%
  10-40%
  20-40%
  20-30%
- Complementarity of Shine-Dalgarno (S-D) sequence with 16s rRNA can affect the
  Rate of transcription
  Rate of translation
  Rate of enzyme
  Rate of cell cycle
- Composition of triplet immediately preceding the AUG ----also affects the efficiency of translation
  Start codon
  Stop codon
  Termination site
- The rate of synthesis of a particular protein will depend on the ---------in the cell
  Steady-state of RNA
  Steady-state of DNA
  Steady-state of mRNA
  Mobile-state of mRNA
- ---------usually proceeds by a combination of endonuclease and 3'exonuclease attack
  Synthesis of mRNA
  Degradation of mRNA
  Degradation of DNA
  Degradation of protein
- One way of increasing the expression of a cloned gene is to increase the
  Temperature

Plasmid copy number
Enzyme
pH

- The loss of plasmids due to defective partitioning is called
  Segregative stability
  Segregative instability
  Non-Segregative stability
  Non-Segregative instability
- Structural instability of plasmids may arise by deletions or rearrangements of
  RNA

mRNA
DNA
rRNA

- Techniques for DNA sequencing became available in the
  Early 1970s
  Late 1970s
  Early 198s
  Late 1980s
- Maxam-Gilbert method for DNA sequencing makes use of chemical reagents to bring about ------ of the DNA
  Base-specific addition
  Base-specific cleavage
  Digestion
  Rearrangement
- DNA synthesis is carried out in the presence of the four deoxynucleoside triphospahtes, one or more of which is labelled with ----- and in four separate incubation mixes
  N^{15}
  P^{32}
  N^{14}
  P^{35}
- The combination of chain terminator and M13 vectors to produce -----is very powerful
  dsDNA
  ssDNA
  dsRNA
  ssRNA
- Sequences that were read beyond 4oo bp contained an average of ---error
  2.3%
  3.2%
  3.4%
  4.4%
- Sequences that were read less than 4oo bp contained an average of ---error
  2.3%
  2.8%
  3.2%

4.4%

- --------is a process to change the genetic information of an organism
  Hybridization
  Autoradiography
  Mutagenesis
  Genetic library
- Mutagenesis sources
  Occur naturally
  Exposure to mutagens
  Induced in laboratory
  All of above
- ----different methods of site-directed mutagenesis
  2

3
4
5

- A synthetic DNA fragment containing the desired mutant sequence is used to replace the corresponding sequence in the wild-type gene
  Cassette mutagenesis
  Primer extension
  Procedures based on PCR
- In -----method, the product of the first PCR is used for the second PCR
  Site directed PCR
  Megaprimer
  Basic PCR
- The ----of a PCR based mutagenic protocol is that the desired mutation is obtained with 100% efficiency
  Advantage
  Disadvantage

  None
- PCR can also be used to generate DNA fragments for gene cloning by using ---with specific restriction sites
  Probes
  Primers
  Both
  None
- Nucleotides are building block of
  Proteins
  Lipids
  DNA
  All of above
- Taq polymerase is a ----DNA
  Thermostable

  Thermoinstable
  Binding resistant
- ----------is a short strand of oligonucleotide that serves as a starting point for DNA synthesis
  Probes
  Primers
  Both
- --------was designed to amplify RNA sequences (mainly mRNA) through synthesis of cDNA by reverse transcriptase
  RT-PCR
  q PCR
  Nested PCR
  Inverse PCR
- It is used to measure starting amounts of DNA, cDNA, or RNA
  RT-PCR
  q PCR
  Nested PCR

Inverse PCR

- In --------two sets of primers are used in two successive PCRs
  RT-PCR
  q PCR
  Nested PCR
  Inverse PCR
- ----------is used to identify the flanking sequences around genomic inserts
  RT-PCR
  q PCR
  Nested PCR
  Inverse PCR
- Multiplex PCR is used to amplify several different DNA sequences simultaneously
  True
  False
- ---------is a type of PCR reaction but the segments of DNA that are amplified are random
  AFLP
  RAPD
  SNPs
  RFLP
- AFLPs-Amplified fragment length polymorphisms are differences in restriction fragments length caused by
  AFLP
  RAPD
  SNPs
  RFLP
- PCR has applications in the fields of life sciences
  Genetic engineering
  Medical
  Forensic
  All of above
  Both a b
- PCR has been used in gene cloning and ------of genomic libraries
  Sequencing
  Screening
  Construction
  Degradation
- DNA profiling or DNA fingerprinting have --------meaning
  Different
  Same
- DNA profiling or DNA fingerprinting is a forensic technique used to identify individuals by characteristics of their..
  Protein
  RNA
  DNA
  cDNA Genomic library
- Assigning of a specific gene to particular region of a chromosome and determining the location of and relative distances between genes on the chromosome is called

Chromosome jumping
Chromosome walking
Chromosome mapping

- Any identifiable feature of the genome cannot serve as a marker in mapping, but specific
  True
  False
- The landmarks on genome map might include
  Short DNA sequences
  Regulatory sites
  Genes themselves
  All of above
- Genetic linkage maps illustrate the order of genes or genetic markers on a chromosome and the relative distances between those genes
  DNA finger printing
  Physical mapping
  Genetic linkage mapping
- ----------give the DNA base pair distances from one landmark to another
  DNA finger printing
  Physical mapping
  Genetic linkage mapping
- ----------exploits variations in homologous DNA sequences
  AFLP
  RAPD
  SNPs
  RFLP

- Botstein et al. (1980) were the first to recognize that ----that target RFLPs can be used for mapping
  DNA probes
  DNA primers
  RNA primers
- Sequence-tagged sites (STS) are more convenient markers than RFLPs because they do not use RT-PCR
  Southern blotting
  Restriction enzyme
  DNA polymerase
- SNPs are -----base-pair positions in genomic DNA at which different sequence alternatives (alleles) exist in a population
  Single
  Double
  Triple
  Complementry
- Amplified fragment length polymorphisms (AFLP) is a diagnostic fingerprinting technique that detects genomic
  Amplified fragment
  Restriction fragments
  Complemenrtry fragment
- FISH is a cytogenetic method that used ----probes

Scanning
Fluorescent
Radio active

- RH mapping used X-ray breakage of chromosomes to determine the distances between ---as well as their order on the chromosome
  DNA probes
  DNA primers
  DNA markers
- Happy mapping is an ------technique
  In vivo
  In vitro
  None
- The process of determining the exact order of nucleotides within a DNA or RNA molecule is called
  Screening
  Sequencing
  Mapping
- Shotgun sequencing is used for sequencing -------DNA strands
  Short
  Long
  Fragmented
  Digested

- --------- a map of each chromosome of the genome is made before the DNA is split up into fragments for sequencing
  Chromosome sequencing
  In clone-by-clone sequencing
  Physical mapping
  Genetic linkage mapping
- Orthologs are homologous genes in different organisms that encode proteins with the
  Different functions
  Same function
  Non functional
- Genome of bacteria may be of variable in size. For example, it may vary from
  0.39Mb to 8.1 Mb
  0.49 Mb to 9.1 Mb
  0.49Mb to 1.9Mb
- Mitochondrial genomes exhibit an amazing
  Genomic diversity
  Structural diversity
  Functional diversity
  Phenotypic diversity
- ----are used to study the expression of many genes at once
  Comparative genomics
  Microarrays
  Filamentous bodies
- Which is the DNA array used in expression analysis
  Spotted DNA arrays

Printed oligonucleotide chips
Both
None

- The predominate application of DNA microarray has been to measure
  Gene expression levels
  Gene synthesis level
  Genetic difference
- Microarrays have been widely used as ----------genotyping platforms
  AFLP
  RAPD
  SNP
  RFLP
- Phage display is the technology that allows expression of exogenous -----on the surface of phage particles
  Monopeptide
  Dipeptide
  Polypeptides
  Nucleotide

- The most common screening method is based on enriching the phage clones with binding affinity for the target by a process called
  Epitope
  Biopanning
  Microarrays
- Applications of phage display includes
  Determination of protein to protein interactions,
  To determine enzyme specificity
  To generate target specific antibodies
  Both a b
  All of above
- A genetic technique in which one of an organism's gene is made inoperative that is simply called
  Knocked in
  Knocked out
  Knocked down
- ---- refers to a gene manipulation method that involves the insertion of a protein coding cDNA sequence at a particular locus in an organism's chromosome
  Knocked in
  Knocked out
  Knocked down
- Small interfering RNA (siRNA) is the most commonly used (RNAi) tool for inducing ----------coding genes
  Long-term silencing of protein
  Short-term silencing of protein
  Permanent silencing of protein